Problem 1
A logistic regression model predicts the probability of a student passing an exam:

$$P(y = 1|x) = \frac{1}{1 + e^{-(0.5x+0.2)}}$$

Here **x** is hours studied.
a) What is the probability of passing if a student studies 2 hours?
b) What is the decision boundary value of **x** (where probability = 0.5)?

Problem 2
Given a single data point: actual **y = 1**, predicted probability **ŷ = 0.3**.
Calculate the **Log Loss** for this single example.

Problem 3
You train a logistic regression model with weight **w = 2.0** and bias **b = -1.0**.
For a feature vector **x = [1.5, 0.5],**
a) Compute **z = w·x + b**
b) Compute the predicted probability **ŷ = σ(z)**.
c) If actual **y = 1**, should the model update **w** to increase or decrease? Brief some reasoning

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Problem 4
A logistic regression model gives the following probabilities for 5 test samples:
[0.1, 0.4, 0.6, 0.9, 0.45`
Actual labels: [0, 0, 1, 1, 0]
Using threshold = 0.5:
a) Find predicted classes.
b) Calculate accuracy.
c) How many false positives and false negatives?

Problem 5
A logistic regression model is trained on a dataset with 2 features.
The learned parameters are: **w = [1.2, -0.8], b = 0.5.**
a) Write the equation of the decision boundary.
b) A new point has **x1 = 2.0, x2 = 1.5**. Which class is predicted?
c) How far (in terms of z value) is this point from the decision boundary?

Problem 6
You are given a highly imbalanced dataset: 95% negative class (0) and 5% positive class (1).
You train a logistic regression model and get 95% accuracy, but it predicts all samples as class 0.
a) Why is accuracy misleading here?
b) Which evaluation metric(s) would you use instead?

c) How can you adjust the model to detect more positive samples?

Problem 7

Write **pseudocode** (or Python-like logic) for the following functions in logistic regression from scratch (no libraries except numpy for basic ops):

a) sigmoid(z)
b) predict_probability(X, w, b)
c) compute_cost(X, y, w, b)
d) compute_gradients(X, y, w, b)
e) gradient_descent(X, y, w_init, b_init, alpha, iterations)

Focus on the loops and formulas, not exact Python syntax.